Lista 2 - Física

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→ Lista Cap 25 - Exs 9E, 23P, 24P, $\boxed{27P}$, 31P, 34P, 36P, $\boxed{47P, 48P, 49P}$

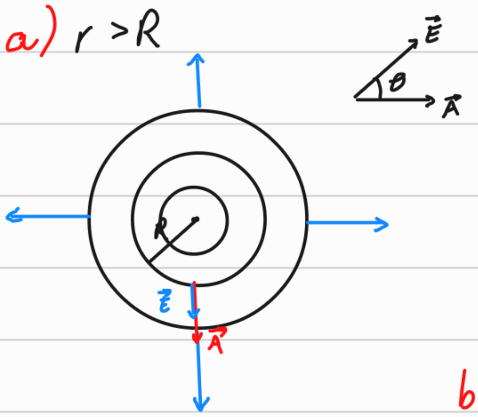
$\Phi = E \cdot A \cdot \cos\theta$; $\Phi = \frac{q_e}{\varepsilon_0}$; $E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{d^2}$; $V = \frac{U}{q}$; $W = -\Delta U$; $V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{d}$

9E - $\Phi = \frac{q_e}{\varepsilon_0}$, $\Phi = E \cdot A \cdot \cos\theta$

$\Phi_T = 6 \cdot \Phi \rightarrow \Phi_T = 6 \cdot \frac{q}{\varepsilon_0}$; $\Phi = \frac{1}{6} \cdot \frac{q}{\varepsilon_0} = \frac{q}{6\varepsilon_0}$ //

23P - $\lambda = \frac{q}{L}$, $E = ?$

a) $r > R$



$\theta = 0$

$\Phi = E \cdot A \cdot \cos\theta \rightarrow \Phi = E \cdot A$

$\Phi = \frac{q}{\varepsilon_0}$ $\frac{q}{\varepsilon_0} = E \cdot 2\pi r L \rightarrow E = \frac{\lambda}{2\pi r \varepsilon_0} \hat{r}$ //

b) $q = 0 \therefore \vec{E} = 0$ //

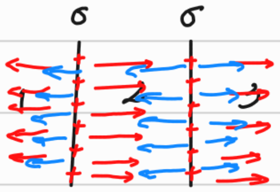
24P -

a) $E = \frac{1}{2\pi\varepsilon_0} \cdot \frac{\lambda}{r} \rightarrow E = \frac{1}{2\pi\varepsilon_0} \cdot \frac{q_e}{Lr}$; $q_e = 0 \rightarrow E = \frac{1}{2\pi\varepsilon_0} \cdot 0 = 0$

b) $q_e = -q \rightarrow E = \frac{1}{2\pi\varepsilon_0} \cdot \frac{-\lambda}{r}$ //

27P-

31P-



$E = \frac{\sigma}{2\varepsilon_0}$ $E = \frac{\sigma}{2\varepsilon_0}$

$E_{(1),(2),(3)} = ?$

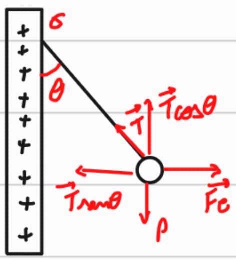
em 2: $E_T = E - E$

$E_{(2)} = 0$ //

em 3 e 1. $E_T = E + E$

$E_T = 2\frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$ //

34P-



$T\cos\theta = p \rightarrow T\cos\theta = m \cdot g$

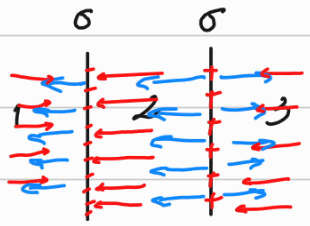
$T sen\theta = qE \rightarrow T = \frac{qE}{sen\theta}$

$E = \frac{\sigma}{2\varepsilon_0}$

$\frac{qE}{sen\theta} \cdot \cos\theta = mg \rightarrow qE = tg\theta\, mg$

$\frac{q\sigma}{2\varepsilon_0} = tg\theta\, mg \rightarrow \sigma = \frac{2\varepsilon_0\, tg\theta\, mg}{q}$ //

36P-



a) $E_{(1)} = 0$ //

b) $E_{(3)} = 0$ //

c) $E_T = E + E$

$E_T = 2\frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$ //

47P-

Cap 26- Exs 34P, 35P, 43E, 56P, 58P, 59P e 60P

; $W = -\Delta U$; $W = -(U_f - U_i)$; $\vec{F_e} = q \cdot \vec{E}$; $dW = F_e(x)\,dx$

$\Delta V = \frac{\Delta U}{q}$; $V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{r}$

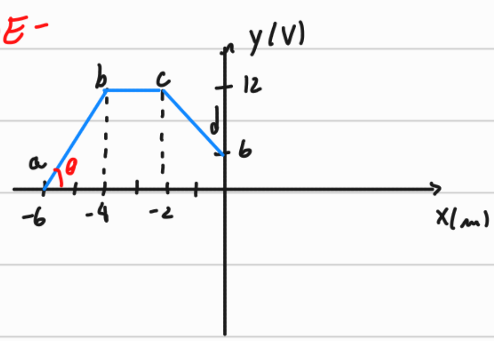
34P- $V = \frac{1}{4\pi\varepsilon_0} \cdot \left(-\frac{5q}{2d} - \cancel{\frac{5q}{d}} + \cancel{\frac{5q}{d}} + \frac{5q}{d}\right)$

$V = \frac{q}{4\pi\varepsilon_0} \cdot \frac{5}{2d} \rightarrow V = \frac{5q}{8\pi\varepsilon_0 d}$ //

35P- $V = \frac{q}{4\pi\varepsilon_0} \cdot \left(\cancel{\frac{+5\cdot 2}{\sqrt{2}d}} - \frac{2\cdot 2}{d} - \cancel{\frac{3\cdot 2}{\sqrt{2}d}} - \cancel{\frac{5\cdot 2}{\sqrt{2}d}} - \frac{2\cdot 2}{d} + \cancel{\frac{3\cdot 2}{\sqrt{2}d}}\right)$

$V = \frac{q}{4\pi\varepsilon_0} \cdot \frac{-8}{d} \rightarrow V = \frac{-8q}{4\pi\varepsilon_0 d}$ //

43E-



$a \rightarrow b \Rightarrow E = ?$

$b \rightarrow c \Rightarrow E = ?$

$c \rightarrow d \Rightarrow E = ?$

$\Delta V = -\int_i^f E(r)\cos\theta\, dr$

$E(x) = -\frac{\partial U}{\partial x}$

$E(x) = -\frac{dV(x)}{dx}$

$E(x) = -tg\,\theta = \frac{\Delta V}{\Delta x}$

$a \rightarrow b$: $E = \frac{-(12-0)}{(-4)-(-6)} = -6\ V/m$

$b \rightarrow c$: $E = \frac{-(12-12)}{-2-(-4)} = 0\ V/m$

$c \rightarrow d$: $E = -\frac{(6-12)}{0-(-2)} = 3\ V/m$

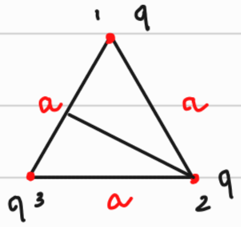
56E - $U = U_{12} + U_{13} + U_{14} + U_{23} + U_{24} + U_{34}$ $\qquad U = q \cdot V$

$$U = \frac{1}{4\pi\varepsilon_0}\left(-\frac{q^2}{a} + \frac{q^2}{a\sqrt{2}} - \frac{q^2}{a} - \frac{q^2}{a} + \frac{q^2}{a\sqrt{2}} - \frac{q^2}{a}\right)$$

$$U = \frac{q^2}{4\pi\varepsilon_0 a} \cdot (-4 + \sqrt{2})$$

//

58P -



$P = 0,31\ kW = 310\ J/s$

$q = 0,12\ C$

$a = 1,7\ m$

$$U = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{a}$$

$$U_i = U_{12} + U_{13} + U_{23}$$

$$U_i = \frac{1}{4\pi\varepsilon_0}\left(\frac{q^2}{a} + \frac{q^2}{a} + \frac{q^2}{a}\right)$$

$$U_i = \frac{3q^2}{4\pi\varepsilon_0 a}$$

$$U_f = U_{12} + U_{23}$$

$$U_f = \frac{1}{\cancel{4}\pi\varepsilon_0}\left(\frac{\cancel{2}q^2}{a} + \frac{\cancel{2}q^2}{a}\right)$$

$$U_f = \frac{q^2}{\pi\varepsilon_0 a}$$

$$\Delta U = U_f - U_i$$

$$\Delta U = \frac{0,12^2}{\pi\varepsilon_0 \cdot 1,7} - \frac{3 \cdot 0,12^2}{4\pi\varepsilon_0 \cdot 1,7}$$

$$\Delta U = 76150588,24\ J$$

$$P = \frac{\Delta U}{t} \rightarrow t = \frac{\Delta U}{P} \rightarrow t \cong 245647\ s$$

$$t \cong 2,84\ \text{dias}$$

//

59P -

$q_1 = -5,0\ \mu C$

$q_2 = +2,0\ \mu C$

$V_{(\infty)} = 0$

a) $V_A = ?$

$$V_A = \frac{1}{4\pi\varepsilon_0}\left(-\frac{q_1}{0{,}15} + \frac{q_2}{0{,}05}\right)$$

$V_A = 6 \cdot 10^4\ V$ //

b) $V_B = ?$

$$V_B = \frac{1}{4\pi\varepsilon_0}\left(\frac{-q_1}{0{,}05} + \frac{q_2}{0{,}15}\right)$$

$V_B = -7{,}8 \cdot 10^5\ V$ //

c) $W = -\Delta U \rightarrow W = U_i - U_f$

$W = q_3 (V_i - V_f)$

$W = q_3 \cdot (6 \cdot 10^4 + 7{,}8 \cdot 10^5)$ → $W = 2{,}5\,J$ //

d) O trabalho aumenta.

e) O trabalho sempre é o mesmo.

60P - $W = U_f - U_i$; $U_i = 0$

$W = U_f$

$$U_{12} = \frac{1}{4\pi\varepsilon_0} \frac{\cancel{4}q \cdot 5q}{\cancel{2}d}$$

$$U_{23} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{-2q \cdot 5q}{d}$$

$U_f = U_{12} + U_{23}$

$$U_f = \frac{1}{4\pi\varepsilon_0}\left(\frac{16 \cdot 10^{-19}}{0{,}014} - \frac{16 \cdot 10^{9}}{0{,}014}\right)$$

$U_f = 0\,J$

$W = 0\,J$ //